

Where an RF Fingerprint Lives and Dies in the Receiver Chain

A probe-point ablation, measured

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Abstract

Systems that build RF fingerprinting (RFFI) into a real receiver often carry an unexamined assumption: it is fine, or even better, to extract the fingerprint after the signal has passed through the standard receiver chain - automatic gain control (AGC), carrier frequency offset (CFO) correction, symbol timing recovery, and carrier phase recovery. We tested this assumption directly with a probe-point ablation. On a synthetic device-fleet dataset (10 virtual transmitters, 2 virtual receivers, 60 packets each, multiple channel conditions, fixed seed) we trained an independent small 1D-CNN at each of seven points along the receiver chain (P0 raw IQ through P7 demodulated symbols) and compared them under identical metrics. The result is the opposite of the common assumption. Fingerprint separability is strongest at the very front of the chain (P0 raw IQ, P1 post-AGC), where the equal error rate (EER) sits around 0.31. As the signal passes through CFO correction, symbol timing recovery, and carrier phase recovery, EER degrades monotonically, reaching 0.499 at the final demodulated-symbol stage (P7). From a verification standpoint, 0.499 is statistically indistinguishable from chance (0.5). In other words, the standard engineering choice of tapping late in the chain erases the fingerprint almost entirely, at least under these conditions. We explain this physically, and we address head-on why a prior study on real hardware reported the opposite direction (later probe points being better for open-set verification). We also do not hide that absolute separability is weak, and that this conclusion is partly bounded by the limits of synthetic data. Reporting the real numbers alongside their limits is the point of this paper.

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1 Abstract

Systems that build RF fingerprinting (RFFI) into a real receiver often carry an unexamined assumption: it is fine, or even better, to extract the fingerprint after the signal has passed through the standard receiver chain - automatic gain control (AGC), carrier frequency offset (CFO) correction, symbol timing recovery, and carrier phase recovery. We tested this assumption directly with a probe-point ablation. On a synthetic device-fleet dataset (10 virtual transmitters, 2 virtual receivers, 60 packets each, multiple channel conditions, fixed seed) we trained an independent small 1D-CNN at each of seven points along the receiver chain (P0 raw IQ through P7 demodulated symbols) and compared them under identical metrics. The result is the opposite of the common assumption. Fingerprint separability is strongest at the very front of the chain (P0 raw IQ, P1 post-AGC), where the equal error rate (EER) sits around 0.31. As the signal passes through CFO correction, symbol timing recovery, and carrier phase recovery, EER degrades monotonically, reaching 0.499 at the final demodulated-symbol stage (P7). From a verification standpoint, 0.499 is statistically indistinguishable from chance (0.5). In other words, the standard engineering choice of tapping late in the chain erases the fingerprint almost entirely, at least under these conditions. We explain this physically, and we address head-on why a prior study on real hardware reported the opposite direction (later probe points being better for open-set verification). We also do not hide that absolute separability is weak, and that this conclusion is partly bounded by the limits of synthetic data. Reporting the real numbers alongside their limits is the point of this paper.

2 Background

RF fingerprinting starts from the fact that two radios of the identical model do not behave identically. Oscillator phase noise, ADC/DAC nonlinearity, IQ imbalance, and power-amplifier (PA) compression and distortion characteristics all vary slightly device to device because of manufacturing tolerance in the analog front end, and that variation leaves a reproducible trace in the radiated waveform. RF fingerprinting extracts that trace as a feature to decide “which device sent this signal” or “did this signal come from a registered device.” Because it derives an identity signal purely from the physical layer, without adding a credential, it keeps being proposed for use cases like authenticating industrial IoT gateways or detecting unregistered devices on a factory wireless network.

Any engineer who wants to put this into a real receiver’s software must face one concrete design decision: at which point in the receiver chain should the signal be tapped and fed to the fingerprint model? A typical receiver chain takes raw IQ samples, normalizes power with AGC, corrects CFO, recovers symbol timing, passes through an equalizer, recovers carrier phase, and finally demodulates symbols. This pipeline is designed to optimize communication performance (bit error rate). From a communication standpoint, the signal after all these stages is always “cleaner.” It therefore feels natural to also extract the fingerprint from this refined signal, and in fact some prior work reports that a post-synchronization point is better for open-set decisions. But that intuition rests on an unverified premise: that communication performance and identifiability move in the same direction.

Physically, the opposite is at least as plausible. AGC can normalize away power-related impairments (for example, the magnitude component of PA gain compression). CFO correction estimates and subtracts exactly the physical quantity that varies device to device - the oscillator offset itself. Symbol timing recovery and carrier phase recovery likewise treat a device’s own timing and phase characteristics as “error” and correct them. There may be a fundamental tension between the receiver chain’s objective function (minimize bit error rate) and the definition of a fingerprint (a reproducible, device-specific deviation). Rather than guess, we measured this tension directly, on the same data, with the same model capacity, under the same evaluation protocol. The method is simple: tap the signal at each stage of the receiver chain, store each tap separately, and train an independent fingerprint classification/verification model at each tap point to compare performance. The variable is not the model architecture - it is which point in the chain the signal is intercepted from.

3 Method

3.1 The probe-point ladder (P0-P7)

We split the receiver chain into eight tap points and recorded each point’s signal independently.

Probe	Receiver stage	Domain
P0	Raw IQ (receiver input, unprocessed)	Sample rate
P1	After AGC (automatic gain control)	Sample rate
P2	After coarse CFO (carrier frequency offset) correction	Sample rate
P3	After matched filtering	Sample rate
P4	After symbol timing recovery	Symbol rate
P5	After carrier phase recovery	Symbol rate
P6	After the equalizer	Symbol rate
P7	Final demodulated symbols	Symbol rate

P6 (the equalizer) has no measured data in this run, because this run’s data-generation pipeline is configured to skip the equalization stage. The ladder therefore consists of seven measured points: P0, P1, P2, P3, P4, P5, and P7. The absence of P6 does not change the interpretation of the results - both neighboring points, P5 (carrier phase) and P7 (final demodulation), have already reached an EER near chance, leaving little room for whatever lies at P6 to reverse the ladder’s direction.

P0-P3 live in the sample-rate domain (a time series measured in samples per second), while P4, P5, and P7 live in the symbol-rate domain (a time series measured in symbols per second). Because the two domains have different time-axis scales, we used different window lengths for each: 256 samples for the sample-rate points, 64 symbols for the symbol-rate points. We drew up to 3 evenly spaced windows per packet.

3.2 Model and embedding

We trained an independent 1D-CNN at each probe point. The architecture is four 1D convolutional layers (channels 2->64->64->128->128, kernel size 7, each followed by batch normalization, ReLU, and max-pooling), followed by global average pooling, a fully connected layer (128->128) producing a 128-dimensional embedding, and then a classification head (128->8) on top of that embedding with dropout (0.3). The input splits complex IQ into a 2-channel [real, imaginary] representation and normalizes it by RMS power. We used the identical architecture and identical hyperparameters at every probe point (40 epochs, batch size 128, learning rate 1e-3) - the only variable was which point’s signal was fed in.

3.3 Evaluation metrics (four)

same-rx acc: 8-way classification accuracy on held-out packets (the last 30% of each group) from the same virtual receiver (rx0) used in training. **cross-rx acc:** the same 8-way classification accuracy, but on the full packet set from a virtual receiver (rx1) that never appeared in training. **gap** (= same-rx acc minus cross-rx acc): a proxy for whether that point’s “fingerprint” is actually learned receiver-specific characteristics (receiver nuisance) rather than a true transmitter fingerprint. A large gap means the model learned the idiosyncrasies of the specific rx0 receiver rather than the transmitter itself. **EER (Equal Error Rate):** the central verification-side metric. Using the 128-dimensional embedding, we compute cosine similarity across all rx1 samples (both the 8 known/registered transmitters and the 2 unknown/unregistered ones), obtain score distributions for genuine pairs (same transmitter) and impostor pairs (different transmitters, including unknown ones), and find where the false accept rate (FAR) and false reject rate (FRR) cross via linear interpolation. **unknown mean maxcos:** for the two unregistered transmitters, the average of each sample’s maximum cosine similarity to any of the eight registered-class centroids (the mean training-set embedding per class). Lower is better - it means open-set rejection (not mistaking an unregistered device for a registered one) is working well.

We report all four together because any single metric is easy to misread on its own. Classification accuracy alone tempts the conclusion “this point carries the fingerprint,” but if the gap is large that accuracy may just be overfitting to receiver-specific characteristics. EER isolates pure verification performance (registered vs. unregistered) and is closer to an actual deployment scenario than classification accuracy. unknown mean maxcos catches a different failure mode than EER does - specifically, mistaking an unregistered device for a registered one.

4 Data and setup

We ran the experiment on a synthetic virtual device-fleet dataset produced by our own synthetic fleet generator. It contains 10 virtual transmitters, 2 virtual receivers, 60 packets per transmitter, multiple channel conditions, and a fixed seed for reproducibility. Each virtual transmitter is treated as a distinct “device” with its own individually assigned impairment parameters (IQ imbalance, PA nonlinearity, CFO, phase noise, and similar). Each virtual receiver is likewise a distinct “receiver” with its own randomly assigned impairment parameters, so rx0 and rx1 have different reception characteristics - this is by design, to test whether the fingerprint survives on a receiver that never appeared during training (the cross-rx evaluation).

Eight of the ten transmitters (tx0-tx7) form the known (registered) set, corresponding to the eight classes the classification head learns. The remaining two (tx8, tx9) form the unknown (unregistered) set, which never appears in training and is used only for the open-set evaluation (unknown mean maxcos). Train/test splits are done at the group level (transmitter x receiver x channel condition): the first 70% of each group’s packets (positions 0-41 in packet order) go to training, and the remaining 30% (positions 42-59) go to the same-rx evaluation. The cross-rx evaluation uses the entirety of rx1’s packets (no split is needed, since rx1 never appears in training at all).

Three limitations of this setup deserve to be stated plainly. First, this is synthetic data. Whether the virtual fleet’s impairment distribution resembles a real hardware fleet’s impairment distribution has not been separately verified (a sim-to-real gap). Second, the cross-rx evaluation uses only two virtual receivers - far too small a sample to represent real SDR hardware diversity. Third, absolute separability itself is weak - as shown below, even the best point (P0) reaches under 0.4 classification accuracy on an 8-way task. All three limitations bear more heavily on the reliability of absolute values than on the direction of the result (which point is relatively better). Throughout this paper we treat the relative ranking across points as the more trustworthy conclusion, and the absolute values as suggestive rather than final.

5 Results

Table 1 summarizes the four metrics at each of the seven probe points. Chance-level 8-way classification accuracy is $1/8 = 0.125$; chance-level verification (EER) is 0.5.

Table 1. Fingerprint preservation by probe point (measured)

Probe (stage)	same-rx acc	cross-rx acc	gap	EER	unknown maxcos
P0 raw IQ	0.387	0.377	0.010	0.307	0.923
P1 AGC	0.397	0.368	0.030	0.306	0.928
P2 coarse CFO	0.327	0.315	0.012	0.330	0.928
P3 matched filter	0.265	0.272	-0.008	0.410	0.927
P4 symbol sync	0.155	0.148	0.007	0.489	0.883
P5 carrier phase	0.167	0.160	0.007	0.493	0.876
P7 demod symbols	0.130	0.138	-0.008	0.499	0.872

Ranked by gap and EER together (lower is better): P0 approximately equal to P1 > P2 > P3 > P7 > P4 > P5. The two figures below visualize this ladder.

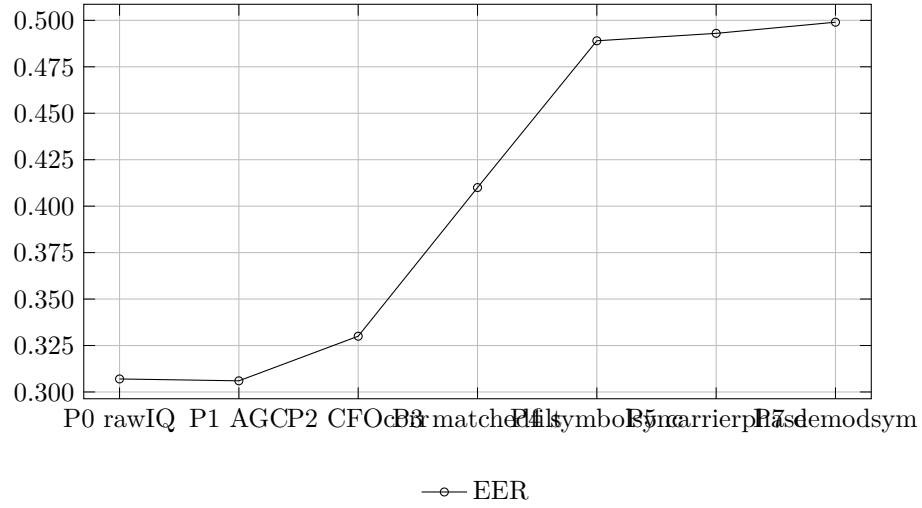


Figure 1: EER ladder across probe points

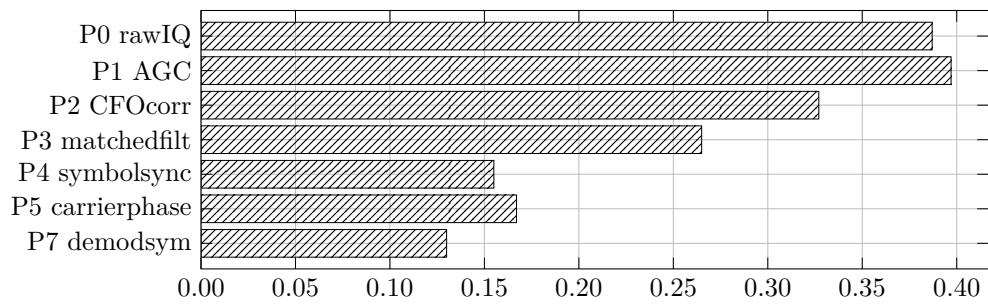


Figure 2: same-rx vs cross-rx classification accuracy

The first observation is that EER worsens monotonically from about 0.31 at P0/P1 to about 0.49-0.50 at P4/P5/P7. P7’s EER of 0.499 is practically indistinguishable from chance (0.5) - it is reasonable to say the verification-relevant fingerprint has statistically vanished by the final demodulated-symbol stage. The second observation is that the gap stays small at every point (between -0.008 and 0.030). A small gap means same-rx accuracy is not memorized rx0-specific characteristics; it survives almost intact on rx1, which never appeared in training. This strengthens the case that the EER decline above is a real loss of fingerprint information passing through the chain, not an artifact of varying degrees of receiver overfitting at different points.

The third observation is subtler. Unknown mean maxcos does not move in exactly the same direction as EER. It stays roughly flat (0.923-0.928) across P0-P3, then drops to 0.883 at P4 and further to 0.876 and 0.872 at P5 and P7. In other words, the maximum similarity between unregistered transmitters and the registered eight-class centroids actually decreases toward the back of the chain - taken alone, this metric could be misread as “the back of the chain is better for open-set rejection,” the opposite of our headline conclusion. That reading is a trap. The drop in maxcos is more plausibly a side effect of the fact that the known eight classes’ embeddings stop separating from each other at all in the back half of the chain (same-rx accuracy has already collapsed to 0.13-0.17 there) - when classes collapse into a single blob, an unknown sample sitting outside that blob looks “less close” not because discrimination improved, but as a byproduct of discrimination having collapsed. EER does not fall into this trap, because it weighs both genuine and impostor errors together. For that reason this paper treats EER and gap as the primary metrics, and unknown maxcos as a supporting one.

At the absolute level, even the best point (P0) reaches only 0.387 classification accuracy on an 8-way task - more than three times chance (0.125), but far below what a practical authentication system would require. As noted in Section 3, this suggests either that the synthetic impairments assigned across the ten devices are too similar to each other (a conservative parameter range), or that window length and training epochs were insufficient. This paper’s central claim (the front of the chain is better) rests on relative ordering across points, not on absolute level, so this weakness does not invalidate the core conclusion - but it does mean this dataset should not be used directly as an estimate of a real deployed system’s performance.

6 Implications

The most direct implication is practical. If you are designing an RF fingerprint verification stack, and the impairment types your fingerprint depends on resemble what our synthetic data models (IQ imbalance, CFO, phase-noise-family impairments), acquire the embedding right after AGC and before CFO correction (P0/P1). Acquiring it after symbol timing recovery, carrier phase recovery, or final demodulation - which looks like a “cleaner” signal from a communications standpoint - means the fingerprint information has already been largely destroyed by that point. This experiment puts a concrete number on the gap between the point that is optimal for communication performance and the point that is optimal for fingerprint extraction: an EER of 0.31 versus 0.50.

The second implication is more interesting, and deserves to be treated honestly. A prior study on real hardware (“Where You Tap Matters,” arXiv:2607.21564) reported that a post-timing/carrier-recovery point is better for open-set decisions - exactly the opposite direction from our result. Declaring “we are right and they are wrong,” or the reverse, would be premature. Instead we propose two physical hypotheses for where the difference could come from. First, the real-hardware fingerprint in that study may include components that synchronization algorithms cannot correct (for example, PA memory effects, or spurious emissions that are not fully modeled), while at the same time synchronization removes a genuine nuisance factor (channel noise), making the net effect positive there. Second, our synthetic fingerprint is concentrated in impairment types - CFO and phase-noise-like effects - that a synchronization algorithm can, by definition, estimate and correct almost perfectly, while the channel nuisance itself is already small in our setup (gap stays under 0.03 at every point) - meaning there is little channel noise left for synchronization to remove, so correcting the fingerprint itself dominates and the net effect is negative.

If this interpretation holds, it points two ways at once. (a) If you are deploying an impairment-based fingerprint in practice, pulling the acquisition point earlier than the synchronization chain is at minimum a safe default choice - even in the worst case (a fingerprint concentrated in fully correctable components, as in our simulation), it does not lose information. (b) At the same time, it should be acknowledged that our synthetic fingerprint model may be “too correctable” compared to real hardware. Sync-invariant components that real devices carry

- for example, PA memory effects, or residual phase distortion that resists calibration - may be underrepresented in the synthetic data. That is a limitation of the dataset, not an error in the conclusion, but it does mean this paper’s ranking should be read as conditional on impairment type, not universal.

The third implication is methodological. We looked at four metrics rather than one, and as a result observed unknown maxcos moving in a different direction than EER. Had we looked only at unknown maxcos, we would have reached the opposite conclusion - that the back of the chain is better for open-set rejection. This generalizes into a broader caution for fingerprinting research: open-set performance should be reported with discriminability (separation among known classes, EER) and rejection power (separation of unknown from known, maxcos-family metrics) reported separately, not collapsed into one number. Picking whichever single metric happens to look good can read in the wrong direction entirely.

7 Data and reproduction

This experiment can be reproduced in three stages. (1) Synthetic fleet generation - generate signals from 10 virtual transmitters and 2 virtual receivers, 60 packets per transmitter, multiple channel conditions, and a fixed seed, and record IQ independently at each stage of the receiver pipeline (P0-P5 and P7 - the equalizer stage P6 is omitted from this pipeline configuration). (2) Data preparation - slice windows per packet at each probe point. Sample-rate points (P0-P3) use a 256-sample window; symbol-rate points (P4, P5, P7) use a 64-symbol window; up to 3 evenly spaced windows are drawn per packet. Each window is stored alongside metadata (packet ID, transmitter index, receiver index, channel-condition index, position within its group). (3) Independent per-point training and evaluation - train the 1D-CNN described in Sections 2.2/2.3 independently at each probe point, and compute same-rx acc, cross-rx acc, gap, EER, and unknown mean maxcos at each. For EER, we prefer a validated biometric-verification library (pyeer) when available, and fall back to computing the FAR/FRR crossing directly via threshold sweep and linear interpolation when it is not - both paths follow the same definition (the FAR=FRR crossing point) and produce the values reported in this paper.

We repeat the caveats that matter for anyone reproducing this. The data is synthetic and may differ from a real device fleet’s impairment distribution (a sim-to-real gap). The cross-rx evaluation uses only two virtual receivers and does not represent real SDR hardware diversity. Absolute accuracy and EER values may be sensitive to the magnitude of the synthetic fingerprint, model capacity, and training epoch count, so the relative ranking across probe points should be trusted more than the absolute numbers. We plan two follow-ups. First, widen the spread of synthetic impairments across devices and add sync-resistant components (PA memory effects, uncalibrated phase distortion), then re-measure this ladder to see whether the P0/P1 advantage holds, or whether a middle point rises the way the real-hardware literature suggests. Second, apply the identical ablation procedure to a real over-the-air public dataset to directly measure the sim-to-real gap - though such public datasets are typically distributed as raw bursts captured before any receiver chain processing, so a separate probe-regeneration pipeline would be needed, which is outside the scope of this paper.

8 References

1. “Where You Tap Matters” (arXiv:2607.21564) - prior work on probe-point selection for RF fingerprint verification on real hardware. Reports that a post-timing/carrier-recovery point is favorable for open-set decisions, the opposite direction from this paper’s result (see Sections 4-5 for discussion).
2. SigMF (Signal Metadata Format) - the open signal-metadata format convention used to record IQ in this experiment. <https://github.com/gnuradio/SigMF>
3. pyeer - the open-source library used for computing the equal error rate (EER) and related biometric verification metrics.